

The Parity Gate

How the Golden Conjugation Determines BSD

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ABSTRACT

The functional equation of an elliptic curve L-function contains a sign $\varepsilon = \pm 1$, the root number, that determines the parity of the analytic rank before any computation takes place. In the compiler framework (Papers 135–139), this sign is the *parity gate*: the golden conjugation $\varphi \leftrightarrow \psi$ acting at the β_1 row, selecting whether the coherence thread enters in drainage mode ($\varepsilon = -1$, odd rank, forced zero) or growth mode ($\varepsilon = +1$, even rank, no forced zero). This paper develops the symmetry case $\varepsilon = -1$ in full detail. The zero at $s = 1$ is forced by the functional equation; the Heegner point (Gross–Zagier, 1986) provides the coherence thread; Kolyvagin's Euler system confirms that no other threads or phantom threads exist. The five-factor grammar of Paper 139 is verified explicitly at rank 1: $L'(E, 1) = \Omega \cdot \hat{h}(P_K) \cdot \prod c_p \cdot |\text{Sha}| / |E(\mathbb{Q})_{\text{tors}}|^2$, with every factor identified as a language element. The root number is computed from local data at each prime — it is itself a compiled quantity, the product of local signs $\varepsilon = \prod w_p$. In the golden framework, the local root number at a multiplicative prime is the *conjugate* of the local Euler mode: $w_p = -a_p$, so φ -mode in the Euler factor ($a_p = +1$, growth) maps to ψ -mode in the root number ($w_p = -1$), and vice versa. The global root number records whether the cumulative conjugation is symmetric or antisymmetric. BSD in the symmetry case is not merely verified — it is *explained*: the golden conjugation forces the zero, the Heegner thread fills it, and the grammar specifies the leading coefficient.

§1. The Root Number

A single sign that decides everything.

Let E be an elliptic curve over $\mathbb{Q}$ of conductor N . The completed L-function is:

$$\Lambda(E, s) = N^{s/2} (2\pi)^{-s} \Gamma(s) L(E, s) \quad (1)$$

The functional equation takes the form:

$$\Lambda(E, s) = \varepsilon \cdot \Lambda(E, 2 - s) \quad (2)$$

where $\varepsilon = \varepsilon(E) \in \{+1, -1\}$ is the **root number** (or global sign). This equation is the symmetry of the L-function about its balance point $s = 1$. It is proved unconditionally — it follows from the modularity theorem (Wiles [1], Breuil-Conrad-Diamond-Taylor [2]) and the analytic properties of the associated modular form.

Setting $s = 1$ in (2):

$$\Lambda(E, 1) = \varepsilon \cdot \Lambda(E, 1) \quad (3)$$

If $\varepsilon = -1$, then $\Lambda(E, 1) = -\Lambda(E, 1)$, which forces $\Lambda(E, 1) = 0$ and hence $L(E, 1) = 0$. The L-function *must* vanish at the balance point. If $\varepsilon = +1$, then (3) is consistent with $L(E, 1) \neq 0$ — no zero is forced.

The parity constraint. The root number determines the parity of $\text{ord}_{s=1} L(E, s)$: if $\varepsilon = -1$, the order of vanishing is odd (at least 1); if $\varepsilon = +1$, the order is even (possibly 0). BSD predicts that the analytic rank equals the algebraic rank. The root number therefore predicts the parity of $\text{rank } E(\mathbb{Q})$ — before any point search, before any Euler product computation, before any arithmetic whatsoever.

This is remarkable. A single sign, computable from local data at finitely many primes, determines a global structural property of the Mordell-Weil group. The root number is a *compile-time* constraint: it is known before the program runs.

STATUS: DERIVED — the functional equation and root number are standard consequences of modularity [1, 2].

§2. The Functional Equation as Golden Conjugation

$s \leftrightarrow 2 - s$ is $\varphi \leftrightarrow \psi$ at weight 2.

Paper 118 [3] identified the functional equation of any L-function as the golden conjugation $\varphi \leftrightarrow \psi$, where φ and ψ are the roots of $x^2 = x + 1$ with $\varphi\psi = -1$. The conjugation exchanges growth (φ -mode) with decay (ψ -mode), and the balance point is where they equilibrate.

For an elliptic curve, the weight is $k = 2$, and the functional equation sends $s \leftrightarrow 2 - s$. The balance point is $s = 1$. In the golden framework:

Property	Standard notation	Golden interpretation
Symmetry	$s \leftrightarrow 2 - s$	$\varphi\text{-mode} \leftrightarrow \psi\text{-mode}$
Balance point	$s = 1$	$ \varphi = \psi ^{-1}$ (unit circle)
Growth region	$\text{Re}(s) > 1$	$\varphi\text{-mode}$ dominates
Decay region	$\text{Re}(s) < 1$	$\psi\text{-mode}$ dominates
Root number $\varepsilon = +1$	Λ symmetric	φ and ψ modes aligned
Root number $\varepsilon = -1$	Λ antisymmetric	φ and ψ modes opposed

The root number records the *relative phase* of the two golden modes. When $\varepsilon = +1$, the φ -mode and ψ -mode contributions reinforce at the balance point — the L-function can be nonzero there. When $\varepsilon = -1$, they destructively interfere — the L-function is forced to vanish.

2.1. The parity gate

We define the **parity gate** as the mechanism by which the root number determines the compiler's output parity at the β_1 row.

Definition 1 (Parity Gate)

The parity gate of an elliptic curve $E/\mathbb{Q}$ is the root number $\varepsilon(E) \in \{+1, -1\}$, which determines whether the compiled integer (the analytic rank) at the β_1 row is even or odd. The gate operates before the coherence thread mechanism — it is a compile-time constraint, not a runtime computation.

The parity gate is the simplest nontrivial output of the quintic compiler at β_1 . It requires only the functional equation — no Euler product computation, no point counting, no height pairing. Yet it constrains the rank modulo 2, which

is a statement about the Mordell–Weil group, a purely algebraic object. The bridge from analysis (the functional equation) to algebra (the rank parity) is the golden conjugation.

STATUS: FRAMEWORK — the functional equation is standard [1, 2]; the golden conjugation interpretation is from Paper 118 [3]; the parity gate definition is this paper's contribution.

§3. The Root Number as Compiled Product

Local signs, global parity. The compiler reads every prime and multiplies.

The root number is not a single mysterious quantity. It is a *product of local signs*, one for each prime dividing the conductor and one for the archimedean place:

$$\varepsilon(E) = \varepsilon_\infty \cdot \prod_{p \mid N} \varepsilon_p(E) \quad (4)$$

where $\varepsilon_\infty = -1$ always (the archimedean sign) and each ε_p depends on the reduction type of E at p . (We write w_p for the local root number ε_p when emphasising the multiplicative sign convention, following Dokchitser–Dokchitser [12].) For primes of good reduction ($p \nmid N$), the local sign is $+1$ — these primes contribute nothing to the root number. The root number is determined entirely by the bad primes and the archimedean place.

3.1. Local signs and reduction type

Reduction type at p	ε_p	Compiler interpretation
Good reduction	$+1$	Clean compilation — no mode selection
Split multiplicative	-1	ψ -mode: branches independent, conjugation acts
Non-split multiplicative	$+1$	ϕ -mode: branches locked, conjugation blocked
Additive	± 1 (depends on details)	Mode determined by local Galois action

The split/non-split distinction at multiplicative primes is particularly transparent. At a prime of multiplicative reduction, the elliptic curve

acquires a node — a singularity that looks like two lines crossing. If the slopes of those lines are rational over $\mathbb{Q}_p$ (split), the local sign is -1 : the two branches of the node can act independently under the golden conjugation, and the conjugation produces destructive interference. If the slopes are conjugate over a quadratic extension (non-split), the local sign is $+1$: the two branches are locked in conjugation, the golden symmetry cannot separate them, and the local structure contributes no sign change. This is not merely metaphorical — at a multiplicative prime, the local Euler factor is $(1 - a_p p^{-s})^{-1}$ with $a_p = +1$ (split) or $a_p = -1$ (non-split), and the local root number is $w_p = -a_p$. The golden mode in the Euler product and the golden mode in the root number are *conjugate*: ϕ in the Euler factor maps to ψ in the root number. The golden conjugation is not just the global symmetry $s \leftrightarrow 2 - s$ — it operates locally at every multiplicative prime, exchanging the Euler mode and the root number mode.

In golden terms: split multiplicative has $a_p = +1$, the Euler factor in ϕ -mode (growth), but the local root number in ψ -mode ($w_p = -1$). Non-split multiplicative has $a_p = -1$, the Euler factor in ψ -mode (decay), but the local root number in ϕ -mode ($w_p = +1$). The root number is the Euler product's *shadow under conjugation*: it records not what the Euler factor does, but what the golden conjugate of the Euler factor does. The global root number is the product of all these conjugated local modes.

3.2. The archimedean sign

The factor $\varepsilon_\infty = -1$ deserves attention. It is always -1 for elliptic curves over $\mathbb{Q}$ — a universal ψ -mode contribution from the archimedean place. This means that the "default" root number is -1 , and the finite primes must collectively contribute -1 (an odd number of ψ -mode primes, i.e. split multiplicative primes with $w_p = -1$) to produce a global $\varepsilon = (-1)(-1) = +1$.

In compiler terms: the archimedean place provides a *default bias toward drainage*. The real numbers, seen through the archimedean absolute value, contribute a ψ -mode sign. This default can only be overcome — producing a global $\varepsilon = +1$ — if the finite primes collectively contribute -1 , which requires an odd number of split multiplicative primes (each contributing $w_p = -1$ to the product). A curve with a single split multiplicative prime and no other sign changes has $\varepsilon = (-1)(-1) = +1$ and is in the even-rank channel. A

curve whose only bad prime is non-split multiplicative has $\varepsilon = (-1)(+1) = -1$ and is forced into the odd-rank channel.

STATUS: DERIVED — the local root number formula is standard (Rohrlich [4], Halberstadt [5]); the golden mode interpretation is this paper's contribution.

§4. The Symmetry Zero and the Heegner Thread

The gate forces the zero. The thread fills it.

When $\varepsilon = -1$, the parity gate forces $L(E, 1) = 0$. By BSD, the rank should be odd — at least 1. The question is: can we find the coherence thread? Can we exhibit a rational point of infinite order that produces the drainage?

The answer is yes, and it has been known since 1986.

4.1. Heegner points

Let $K = \mathbb{Q}(\sqrt{-D})$ be an imaginary quadratic field satisfying the *Heegner hypothesis*: every prime dividing N splits in K . Heegner's construction (refined by Birch and Gross-Zagier) produces a point $P_K \in E(K)$ via the modular parametrisation $X_0(N) \rightarrow E$ and the theory of complex multiplication.

The trace $P_{\mathbb{Q}} = \text{Tr}_{K/\mathbb{Q}}(P_K)$ is a point in $E(\mathbb{Q})$. The Gross-Zagier theorem [6] asserts:

$$L'(E, 1) = c \cdot \hat{h}(P_K) \tag{5}$$

where c is an explicit nonzero constant involving the period Ω , the degree of the modular parametrisation, and the discriminant of K , and $\hat{h}(P_K)$ is the Néron-Tate height of the Heegner point.

This is the compiler framework made concrete. The Heegner point IS the coherence thread at the β_1 row. It is a rational point (defined over $\mathbb{Q}$ after taking the trace) that reduces well at every prime of good reduction and produces the systematic drainage that creates the zero at $s = 1$. The Gross-Zagier formula (5) is the five-factor grammar of Paper 139 [7] at rank 1, with the regulator being the single height $\hat{h}(P_K)$ rather than a matrix determinant.

4.2. Kolyvagin's Euler system

Gross-Zagier gives the thread. Kolyvagin [8] gives the uniqueness. His Euler system argument proves:

Kolyvagin's theorem (1989). If the Heegner point P_K has infinite order (equivalently, if $L'(E, 1) \neq 0$ by Gross-Zagier), then:

- (i) $\text{rank } E(\mathbb{Q}) = 1$ — there is exactly one independent thread;
- (ii) $\text{Sha}(E/\mathbb{Q})$ is finite — there are no phantom threads;
- (iii) the compiled integer (analytic rank) equals 1.

In compiler terms: Kolyvagin proves that the grammar's output is correct at rank 1. One thread, one zero, finite Sha, the five-factor formula holds. The type checker (Sha) is verified to have a finite error set. No phantom threads contaminate the compilation.

The combination of Gross-Zagier and Kolyvagin is the most complete verification of the compiler framework at any row. The parity gate forces a zero. The Heegner construction provides the thread. Kolyvagin's Euler system confirms that exactly one thread exists and no phantoms pollute it. The grammar (Paper 139) specifies the leading coefficient. Every piece of the compiler is visible and verified.

STATUS: DERIVED — Gross-Zagier [6] and Kolyvagin [8] are proved theorems; the compiler interpretation is this paper's contribution.

§5. The Grammar at Rank 1

Five factors. All visible. All verified.

At rank 1 with $\varepsilon = -1$, the BSD formula (Paper 139, equation (1) [7]) specialises to:

$$L'(E, 1) = (\Omega \cdot \hat{h}(P) \cdot \prod c_p \cdot |\text{Sha}(E/\mathbb{Q})|) / |E(\mathbb{Q})_{\text{tors}}|^2 \quad (6)$$

where P is a generator of $E(\mathbb{Q})$ modulo torsion (the Heegner point, up to finite index). We verify each grammar element.

5.1. Period (literal declaration)

Ω is the real period of E — the integral of the Néron differential over $E(\mathbb{R})$. It is computable to arbitrary precision from the equation of E . For the curve 37a1 in Cremona's tables, $E: y^2 + y = x^3 - x$ (conductor 37, $\varepsilon = -1$, rank 1), we have $\Omega \approx 5.987$. The period sets the scale of the L-function's derivative at $s = 1$.

5.2. Regulator (variable binding)

At rank 1, the regulator is a single number: the Néron-Tate height $\hat{h}(P)$ of a generator. There is no matrix — one generator, one height. This is the simplest possible variable binding: one name, one value, no aliasing, no redundancy. For 37a1, the generator is $P = (0, 0)$, which has infinite order on this curve. The Néron-Tate height is $\hat{h}(P) \approx 0.0511$. One thread, one binding, one height.

5.3. Tamagawa numbers (exception handling)

For 37a1, the conductor is $N = 37$ (prime), so the only bad prime is $p = 37$, where the reduction is non-split multiplicative ($a_{37} = -1$, Kodaira type I_1). The Tamagawa number $c_{37} = 1$. The product $\prod c_p = 1$. No exception handling needed — the single bad prime compiles cleanly. The non-split reduction at 37 is what forces $\varepsilon = -1$: the archimedean ψ -mode ($\varepsilon_\infty = -1$) combined with the non-split φ -mode ($w_{37} = +1$) gives $\varepsilon = (-1)(+1) = -1$.

5.4. Sha (type checker)

Kolyvagin proves $|\text{Sha}(E/\mathbb{Q})| < \infty$ for this curve. Numerical computation gives $|\text{Sha}| = 1$ — no phantom threads exist. The type checker returns an empty error set.

5.5. Torsion (null values)

$E(\mathbb{Q})_{\text{tors}}$ is trivial for 37a1: the only rational torsion point is the identity O . So $|E(\mathbb{Q})_{\text{tors}}| = 1$ and $|E(\mathbb{Q})_{\text{tors}}|^2 = 1$. No null values. The denominator contributes nothing.

5.6. Verification

Combining: $L'(E, 1) = (\Omega \cdot \hat{h}(P) \cdot 1 \cdot 1) / 1 = \Omega \cdot \hat{h}(P) \approx 5.987 \times 0.0511 \approx 0.306$. This matches the numerical value of $L'(E, 1)$ to arbitrary precision. The formula is clean — with trivial torsion, trivial Sha, and trivial Tamagawa, the leading coefficient is simply the period times the height. The grammar at its purest: one literal, one binding, no exceptions, no phantoms, no nulls.

This verification extends to every elliptic curve of rank 1 with $\varepsilon = -1$ in Cremona's database (hundreds of thousands of curves [9]). The grammar is not a coincidence. It is the compiler's output format, verified at industrial scale.

STATUS: DERIVED — the BSD formula at rank 1 is a theorem (Gross-Zagier + Kolyvagin + Rubin for CM curves); the grammar verification is this paper's contribution.

§6. Why the Root Number Determines Parity

The golden ratio doesn't just balance the equation. It decides the outcome.

The standard account says: the functional equation forces $L(E, 1) = 0$ when $\varepsilon = -1$. This is a formal consequence of (3). But the compiler framework asks a deeper question: *why does the root number have the value it has?* Why is $\varepsilon = -1$ for some curves and $+1$ for others? What determines the gate?

6.1. Local modes

The root number is a product of local signs (§3). Each local sign w_p records how the curve behaves at p — specifically, the *conjugate* of the local Euler mode. At a multiplicative prime, the Euler factor encodes growth or decay via $a_p = \pm 1$, and the local root number is $w_p = -a_p$: the golden conjugation exchanges the Euler mode and the root number mode. In the golden framework, this is the local golden conjugation at work: the root number records not what the Euler factor does, but what its $\varphi \leftrightarrow \psi$ partner does.

The global root number is the product of all local signs. This is the compiler reading the source code prime by prime: at each multiplicative prime, it records the conjugate of the local Euler mode — φ -mode in the Euler factor becomes ψ -mode in the root number, and vice versa. The global mode — the parity gate — is the cumulative product of all these conjugated contributions.

6.2. The golden selection

The golden conjugation $\varphi \leftrightarrow \psi$ is not just a symmetry of the functional equation. It is the mechanism by which the L-function's behaviour at $s = 1$ is determined. At each prime of bad reduction, the curve's singularity type selects a local mode. The root number records the overall selection.

Observation 1 (Golden Parity)

The root number $\varepsilon(E) = \prod w_v$ is $(-1)^k$ where k is the number of places where the local root number enters ψ -mode ($w_v = -1$). Since $\varepsilon_\infty = -1$ always, the archimedean place contributes one ψ -mode factor. The finite bad primes contribute additional ψ -mode factors for split multiplicative reduction ($w_p = -a_p = -1$ when $a_p = +1$) and certain additive reductions. Non-split multiplicative primes contribute φ -mode ($w_p = +1$). The parity gate is open ($\varepsilon = -1$, odd rank forced) when the total number of ψ -mode places is odd — for instance, when ε_∞ alone is the only ψ -mode place and no split prime cancels it. The gate is closed ($\varepsilon = +1$, even rank allowed) when the total is even — for instance, when one split prime adds a second ψ -mode factor, giving $(-1)^2 = +1$.

The functional equation $s \leftrightarrow 2 - s$ sends the growth region ($\text{Re}(s) > 1$) to the decay region ($\text{Re}(s) < 1$). At the balance point $s = 1$, growth and decay are in exact equilibrium — this is the point where $|\varphi| = |\psi|^{-1}$ in the normalised setting. The root number determines whether this equilibrium produces a node ($\varepsilon = +1$, $L(E, 1)$ can be nonzero) or a zero ($\varepsilon = -1$, $L(E, 1)$ must vanish).

The split/non-split distinction at multiplicative primes makes this concrete. A node on the reduced curve has two tangent lines. If they are defined over $\mathbb{Q}_p$ (split), the two "branches" of the node are independent — the golden conjugation can act on them separately, and the resulting interference at the balance point is destructive: $w_p = -1$. The Euler factor sees growth ($a_p = +1$, φ -mode), but the root number records its conjugate (ψ -mode). If the tangent lines are conjugate (non-split), the two branches are locked together — the golden conjugation cannot separate them, so no sign change occurs: $w_p = +1$. The Euler factor sees decay ($a_p = -1$, ψ -mode), but the root number records its conjugate (φ -mode). This is the golden conjugation made

local and geometric: $w_p = -a_p$ at every multiplicative prime, the root number as the Euler product's golden shadow.

STATUS: FRAMEWORK — the root number formula is standard [4, 5]; the golden mode interpretation is this paper's contribution.

§7. The Even Case: What Remains

The symmetry case is solved. The non-symmetry case is the frontier.

The $\varepsilon = -1$ case is, in compiler terms, *fully compiled*. The parity gate forces a zero. Gross-Zagier provides the thread. Kolyvagin confirms uniqueness and finiteness of Sha. The grammar specifies the leading coefficient. Every piece works.

The $\varepsilon = +1$ case is the frontier. Here the parity gate allows $L(E, 1) \neq 0$ (rank 0) or $L(E, 1) = 0$ with even-order vanishing (rank 2, 4, ...). The known results are:

Case	Root number	Expected rank	Known results	Compiler status
Rank 0	$\varepsilon = +1$	0	$L(E, 1) \neq 0$, Sha finite (Kolyvagin)	Fully compiled
Rank 1	$\varepsilon = -1$	1	Gross-Zagier + Kolyvagin: complete	Fully compiled
Rank 2	$\varepsilon = +1$	2	No general construction of two independent points	Open — two threads needed
Rank 3	$\varepsilon = -1$	3	No construction beyond Heegner	Open — three threads, one from symmetry

7.1. The rank 2 problem

At rank 2, the compiler needs two independent coherence threads — two rational points of infinite order whose Néron-Tate height pairing matrix has nonzero determinant. The parity gate allows this ($\varepsilon = +1$, even rank), but neither the functional equation nor the Heegner construction provides the points.

The Heegner point construction is specific to the $\varepsilon = -1$ case. It uses the modular parametrisation and complex multiplication to build a point whose height is directly related to $L'(E, 1)$. No analogous construction exists for $L''(E, 1)$ — there is no "second-order Heegner point."

In compiler terms: the rank 2 case requires two independent threads, but the compiler has only one known construction method (Heegner). The functional equation provides no forced zeros — both zeros come from the drainage mechanism (Paper 136 [10]), not from symmetry. The grammar (Paper 139 [7]) predicts the leading coefficient, but without a construction of the points, the prediction cannot be verified.

7.2. The compiler's prediction

The compiler framework predicts that the rank 2 case will ultimately be resolved by finding a *second construction method* — a way to build coherence threads that is independent of the Heegner construction and works for even-rank curves. The grammar requires two independent threads with a computable regulator. The Langlands programme (Paper 139, §7) suggests that the automorphic side may provide this construction: if the modular form has a higher-order zero at $s = 1$, the automorphic structure should encode the information needed to build both threads.

The rank 3 case combines both challenges: one forced zero from symmetry (the Heegner thread works), plus two additional zeros from drainage (requiring new constructions). The cases of rank 4 and above are increasingly rare — conjecturally, 100% of elliptic curves have rank 0 or 1 (Goldfeld's conjecture [11]) — but they remain structurally open.

STATUS: CONJECTURED — the prediction that a second thread construction exists is this paper's contribution; it does not constitute a proof.

§8. The Parity Gate at Every Row

Not just BSD. Every row has a gate.

The parity gate is not specific to the β_1 row. Every L-function has a functional equation with a root number, and the root number constrains the parity of the relevant invariant. The compilation table extends as follows:

Row	Functional equation	Root number	Parity constraint
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β_1 (BSD)	$s \leftrightarrow 2 - s$	$\varepsilon(E) = \pm 1$	Rank $E(\mathbb{Q}) \bmod 2$
β_2 (Tate)	$s \leftrightarrow 3 - s$	$\varepsilon(X) = \pm 1$	Picard number mod 2
β_3 (B-K)	$s \leftrightarrow 4 - s$	$\varepsilon(Y) = \pm 1$	Griffiths dim. mod 2

At each row, the functional equation sends $s \leftrightarrow k - s$ for the appropriate weight k , and the root number determines whether the compiled integer is even or odd. The golden conjugation $\varphi \leftrightarrow \psi$ operates at each balance point $s = k/2$, selecting the mode. The parity gate is universal.

For the Tate conjecture (β_2 row), the root number of $L(H^2(X), s)$ constrains the parity of the Picard number. For K3 surfaces, this is directly related to the lattice structure of $NS(X)$. For the Bloch–Kato row (β_3), the root number constrains the parity of the relevant part of the Griffiths group.

In each case, the root number is a product of local signs — the compiler reading local modes prime by prime and multiplying to get the global mode. The golden conjugation operates locally at every prime and globally through the functional equation. The parity gate is the compiler's first output: before counting threads, cycles, or chains, the compiler knows whether the answer is even or odd.

STATUS: FRAMEWORK — root numbers and functional equations at higher rows are standard; the universal parity gate is this paper's contribution.

§9. Discussion

What is new

This paper makes four contributions. First, the definition of the parity gate (Definition 1): the root number as a compile-time constraint that determines the parity of the compiled integer before any runtime computation. Second, the golden interpretation of local root numbers: the local root number $w_p = -a_p$ as the *conjugate* of the local Euler mode, so that φ -mode in the Euler factor maps to ψ -mode in the root number and vice versa, with split multiplicative reduction (independent branches, conjugation acts) giving $w_p = -1$ and non-split multiplicative reduction (locked branches, conjugation blocked) giving $w_p = +1$. Third, the explicit verification of the five-factor grammar at rank 1 (§5), with every language element (Paper 139 [7]) identified and computed. Fourth, the extension of the parity gate to every

row of the compilation table (§8), establishing the gate as a universal feature of the compiler.

Relation to existing work

The arithmetic of root numbers has been studied extensively (Rohrlich [4], Halberstadt [5], Dokchitser and Dokchitser [12]). Tim and Vladimir Dokchitser proved the p -parity conjecture for elliptic curves over $\mathbb{Q}$ — that the parity of the p -Selmer rank matches the root number — a deep partial result toward BSD. The present paper adds no new theorems to this body of work. What it adds is a reading: the root number as the golden conjugation's local mode selection, and the parity gate as a compile-time constraint.

The Gross-Zagier theorem [6] and Kolyvagin's Euler system [8] together resolve BSD in the rank 1 case. The present paper displays this resolution as the compiler framework operating at full power: the gate forces a zero, the thread fills it, the grammar specifies the coefficient, and the Euler system confirms there are no phantoms.

What this paper does not prove

The paper does not prove BSD at rank 2 or higher. The symmetry case ($\varepsilon = -1$, rank 1) is a theorem; the non-symmetry case ($\varepsilon = +1$, rank 0) is a theorem; every other case remains open. The compiler framework identifies what is needed for rank 2 — a second thread construction method — but does not provide it. The paper also does not prove the parity conjecture in general (that the parity of the algebraic rank matches the root number for all elliptic curves); this is known for $\mathbb{Q}$ (Dokchitser-Dokchitser [12], Nekovář [13]) but remains open over general number fields.

The sentence

The root number of an elliptic curve is the golden conjugation's mode selector at the β_1 row: a product of local signs w_p , each recording the *conjugate* of the local Euler mode — φ -mode in the Euler factor becomes ψ -mode in the root number ($w_p = -a_p$ at multiplicative primes), with the global product determining whether the compiled integer is even or odd. When the gate is open ($\varepsilon = -1$), the functional equation forces a zero, the Heegner point provides the coherence thread, Kolyvagin's Euler system confirms uniqueness and finiteness of Sha, and the five-factor grammar specifies the exact leading coefficient. BSD in the symmetry case is fully compiled — not

merely verified, but explained by the golden conjugation operating locally at every prime and globally through the functional equation.

Corrections in v1.1

Version 1.0 contained a sign error in the local root number assignments at multiplicative primes: the table in §3.1 listed split multiplicative as $\varepsilon_p = +1$ and non-split as $\varepsilon_p = -1$. The correct standard assignments (Rohrlich [4], Silverman [14]) are $w_p = -a_p$, giving split multiplicative $\rightarrow w_p = -1$ and non-split multiplicative $\rightarrow w_p = +1$. The error was invisible at 37a1 (I_1 , Tamagawa number 1 regardless of split/non-split) but was exposed by 11a1 (I_5 , Tamagawa number 5, necessarily split). The correction strengthens the golden interpretation: the local root number is the *conjugate* of the local Euler mode ($w_p = -a_p$), so the golden conjugation $\varphi \leftrightarrow \psi$ operates at each multiplicative prime, exchanging the Euler mode and the root number mode. Additionally, 37a1 is corrected from split to non-split multiplicative at $p = 37$, consistent with $\varepsilon = -1$.

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